

SORTAVALA, Russian Federation

WMO: 228020

Lat: 61.7140N

Lon: 30.7133E

Elev: 62

StdP: 14.66

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.7	-8.8	-19.5	1.9	-14.5	-13.3	2.7	-8.5	19.6	26.8	17.6	26.7	1.9	320	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	79.2	66.2	76.1	64.5	73.0	62.6	68.7	75.9	66.5	73.0	64.5	71.0	5.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	95.9	72.5	63.8	88.9	70.2	61.8	82.6	68.1	32.9	76.0	31.1	73.1	29.7	71.0	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	13.9	12.0	DB	-19.6	83.0	7.2	3.8	-24.8	85.8	-29.0	88.1	-33.0	90.2	-38.2	93.0	(4)
(5)			WB	-19.8	70.9	7.1	2.8	-25.0	72.9	-29.1	74.6	-33.1	76.2	-38.3	78.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.6	18.4	17.8	25.7	36.9	48.1	56.9	63.3	59.8	50.8	40.4	31.2	23.9
	DBStd	17.79	13.29	12.85	8.72	6.69	7.36	6.17	5.59	5.26	5.93	6.81	8.22	11.92
	HDD50	4903	980	902	754	397	124	10	0	1	61	302	564	808
	HDD65	9361	1445	1322	1219	844	524	254	99	177	427	763	1014	1273
	CDD50	1092	0	0	0	3	66	217	411	305	85	5	0	0
	CDD65	73	0	0	0	0	1	11	45	16	0	0	0	0
	CDH74	498	0	0	0	0	14	99	291	94	0	0	0	0
	CDH80	59	0	0	0	0	0	14	41	4	0	0	0	0
Wind	WSAvg	5.1	5.4	5.3	5.4	4.9	5.3	4.9	4.6	4.4	4.7	5.2	5.9	5.7
Precipitation	PrecAvg	24.9	1.7	1.3	1.2	1.3	1.5	2.2	2.4	3.1	2.6	2.6	2.6	2.4
	PrecMax	35.1	4.5	4.0	2.6	2.6	4.7	5.4	7.0	6.1	5.3	5.7	4.9	5.6
	PrecMin	16.1	0.3	0.2	0.2	0.2	0.1	0.4	0.4	0.9	0.6	0.4	0.9	0.5
	PrecStd	4.7	0.9	0.7	0.6	0.6	0.9	1.1	1.3	1.3	1.3	1.5	1.0	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.8	39.5	48.2	62.8	75.9	81.9	84.2	80.2	70.4	56.7	48.2	44.6
		MCWB	36.6	36.2	39.9	51.8	59.4	68.0	69.8	67.8	61.2	53.1	46.4	42.5
	2%	DB	36.8	36.8	42.8	57.1	70.5	76.7	80.4	76.8	65.4	53.9	45.9	41.1
		MCWB	35.5	34.9	36.9	47.8	56.9	63.5	67.2	65.1	58.6	51.5	44.5	39.8
	5%	DB	35.3	35.0	38.8	52.2	65.8	72.8	77.2	73.3	62.5	52.1	44.0	38.7
		MCWB	34.2	33.6	34.4	44.2	53.8	61.2	66.0	63.0	56.8	49.8	42.8	37.5
	10%	DB	33.7	33.6	36.4	47.9	61.6	69.2	74.2	70.1	59.9	50.2	42.0	36.8
		MCWB	32.9	32.6	33.4	41.2	51.6	59.1	64.1	62.0	55.6	48.3	40.9	35.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	36.9	40.7	52.6	62.9	69.9	72.6	70.3	62.3	54.3	47.0	42.8
		MCDB	38.2	38.2	46.6	60.1	71.1	79.8	81.3	76.7	66.5	55.8	47.8	44.1
	2%	WB	35.7	35.1	37.6	48.6	58.4	66.1	69.6	66.8	59.8	52.3	44.7	40.1
		MCDB	36.5	36.4	41.6	55.5	66.5	74.2	77.0	73.6	63.7	53.7	45.6	41.0
	5%	WB	34.3	33.9	35.4	45.2	55.4	62.9	67.6	64.8	58.0	50.4	43.0	37.6
		MCDB	35.1	34.8	37.7	51.6	64.4	70.1	74.3	71.4	61.1	51.7	43.9	38.6
	10%	WB	33.0	32.7	34.0	41.7	52.6	60.3	65.6	62.8	56.4	48.4	41.0	35.7
		MCDB	33.8	33.6	35.7	46.9	60.1	67.6	71.6	68.4	59.4	49.8	42.0	36.7
Mean Daily Temperature Range	5% DB	MDBR	9.6	11.4	13.4	14.8	16.6	15.5	14.4	14.5	12.0	8.5	6.8	8.3
		MCDBR	6.4	8.4	15.0	22.5	23.7	21.3	19.2	19.9	15.8	10.1	7.2	6.3
		MCWBR	6.1	7.9	10.7	13.6	12.6	10.2	8.7	9.5	9.4	7.8	7.1	6.1
	5% WB	MCDBR	6.6	7.8	11.8	21.1	20.4	17.8	16.5	16.6	13.0	8.7	6.8	5.8
		MCWBR	6.5	7.4	8.9	13.4	12.1	10.0	8.8	9.0	8.6	7.4	6.9	5.8
Clear-Sky Solar Irradiance	taub	0.250	0.297	0.323	0.361	0.359	0.352	0.381	0.366	0.341	0.322	0.270	0.220	
	taud	2.036	2.093	2.176	2.316	2.438	2.486	2.422	2.448	2.503	2.396	2.126	1.977	
	Ebn at Noon	136	205	247	262	274	278	265	259	246	203	128	89	
	Edh at Noon	12	23	30	33	31	31	32	29	23	18	11	7	
All-Sky Solar Radiation	RadAvg	67	245	712	1173	1603	1723	1664	1274	726	283	71	34	
	RadStd	10	61	98	124	162	105	145	145	91	40	12	4	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.03	N/A	N/A	N/A	N/A	N/A	-385	N/A	N/A	N/A
Regional (20 neighbors)		+0.97	N/A	N/A	N/A	N/A	N/A	-360	N/A	N/A	N/A

Nomenclature: See separate page